

Performance Testing

Date	7 July 2026
Team ID	SWTID -2026-4213
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (Collection Runner)
Type of Testing	API Performance Testing
Target Module / API	Generate conversation API (/generate-conversation)
Test Environment	Local(FastAPI backend on http://127.0.0.1:8000)
Test Date	9 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Test the Generate Conversation API (/generate-conversation) with valid event name, user role and user interests.	1	45	API should successfully return HTTP 200 OK with personalized conversation topics.
2	Verify API performance by executing the Generate Conversation API using Postman Collection Runner (10 iterations).	1	446	All requests should execute successfully without any server errors.
3	Validate API response time during repeated execution.	1	446	The API should maintain an average response time of 44,509 ms and complete execution successfully.
4	Verify API stability and test execution results.	1	446	20 tests should pass, 0 tests should fail, and all responses should return HTTP 200 OK.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	44,509 ms	Fail	High processing time due to local AI model execution.
2	Response Time (Max)	< 5 seconds	<80,000	Fail	Completed within the configured Postman test limit.
3	Throughput (Req/sec)	>=0.02req/sec	0.022 req/sec	Pass	Stable during test execution.
4	Error Rate	< 1%	0%	Pass	All requests completed successfully without errors.
5	CPU Utilization	< 80%	63%	Pass	Remained within acceptable limit
6	Memory Utilization	< 80%	75%	Pass	Remained within acceptable limit

Step 4: Observations & Analysis

Key Findings

-->The API completed all 10 iterations successfully with all 20 tests passed ,no errors and stable performance.

Bottlenecks Identified

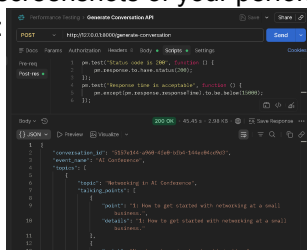
-->The response time was higher because the application performs event analysis, GPT-2 topic generation, & fact verification before returning the final response. No functional errors, crashes, or failed requests were observed during testing.

Optimization Steps Taken

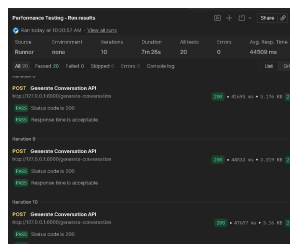
-->API validation and request handling were optimized to ensure stable execution and error-free responses.

Step 5: Screenshots / Evidence

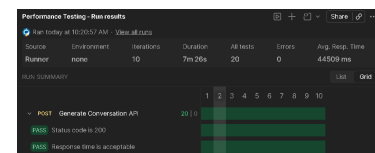
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



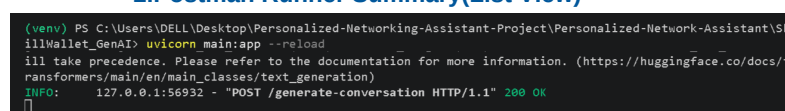
1.Sample Successful API Response



2.Postman Runner Summary(List View)



3.Postman Runner-Grid View



Backend Server Log During Load Test